LEARNER GUIDE

Repeated change: compound interest and depreciation

Percentages, Ratio, Growth and Bounds · Lesson 6. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Simple interest adds the same amount every year: a percentage of the original. **Compound interest** adds a percentage of the amount so far, so interest earns interest. Each year multiplies by the same multiplier, so after n years the amount is the original × multiplier^n. **Depreciation** works the same way with a multiplier below 1: losing 15% a year multiplies by 0.85 each year.

What you are learning to notice

- Simple interest grows by the same amount every year.
- Compound interest grows by more each year: the extra is interest on interest.
- Repeated percentages multiply. Two years of 5% is × 1.05^2 = × 1.1025, not + 10%.
- Depreciation by a percentage never reaches £0 (unless it is 100%).
- Money is given to the nearest penny; say so when you round.

Moving through it, one step at a time

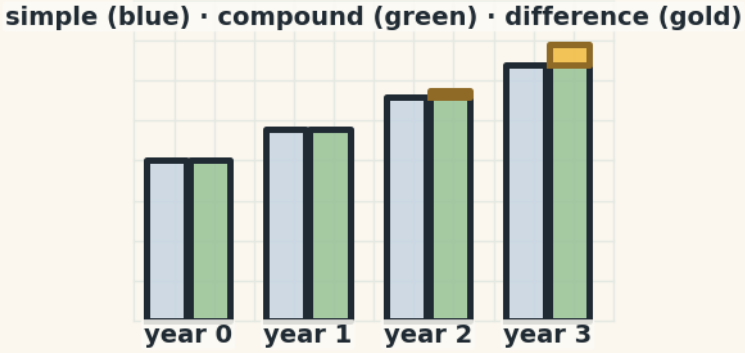
Write the multiplier for one year (5% interest: 1.05; 25% depreciation: 0.75). Multiply the original by it once for every year, or use a power: original × multiplier^n. For simple interest, add the same amount (the rate × the original) each year.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Compound interest

£400 is saved at 20% compound interest for 3 years. How much is there? How does it compare with simple interest?

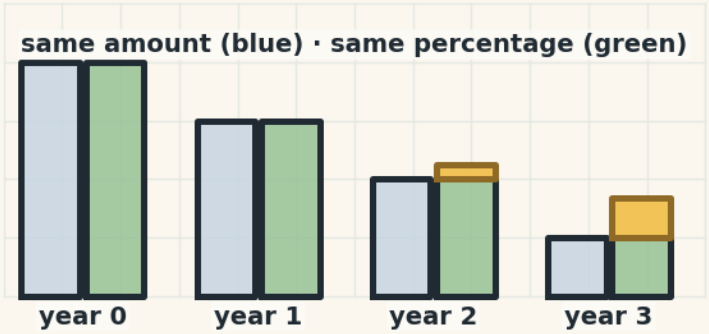


1. The multiplier for one year	100% + 20% = 120% = 1.2
2. Year by year	400 × 1.2 = 480; 480 × 1.2 = 576; 576 × 1.2 = 691.20
3. As a power	$400 \times 1.2^3 = 400 \times 1.728 = 691.20$
4. Simple interest	400 + 3 × 80 = 640

£691.20 with compound interest: £51.20 more than simple interest, because of interest on interest.

Example 2 — Depreciation

A bike worth £800 depreciates by 25% a year. What is it worth after 3 years?



1. The multiplier for one year	100% – 25% = 75% = 0.75
2. Year by year	800 × 0.75 = 600; 600 × 0.75 = 450; 450 × 0.75 = 337.50
3. As a power	$800 \times 0.75^3 = 800 \times 0.421875 = 337.50$

£337.50. (Losing £200, 25% of the original, every year would leave only £200.)

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (percentages as multipliers (Lesson 3) and powers), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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